

Aim 1 - Sanchez-Vega pathways by brain-met prevalence (cohort=genie; brain-met n=393, no-brain-met n=841); Fisher exact, Holm and BH-FDR across 10 tests

Pathway	Mut in brain-met (n/N)	% brain-met	Mut in no-brain-met (n/N)	% no-brain-met	Prev. ratio (95% CI)	Fisher p	Holm p	BH q
p53	250/393	63.6%	459/841	54.6%	1.17 (1.06-1.28)	0.0030	0.0270	0.0150
RTK/RAS	248/393	63.1%	448/841	53.3%	1.18 (1.07-1.31)	0.0013	0.0134	0.0134
PI3K	197/393	50.1%	420/841	49.9%	1.00 (0.89-1.13)	1.0000	1.0000	1.0000
Cell cycle	150/393	38.2%	273/841	32.5%	1.18 (1.00-1.38)	0.0535	0.3747	0.1337
Notch	90/393	22.9%	155/841	18.4%	1.24 (0.99-1.56)	0.0779	0.4676	0.1559
Myc	63/393	16.0%	124/841	14.7%	1.09 (0.82-1.44)	0.5521	1.0000	0.7887
Wnt	42/393	10.7%	61/841	7.3%	1.47 (1.01-2.14)	0.0468	0.3747	0.1337
Hippo	26/393	6.6%	58/841	6.9%	0.96 (0.61-1.50)	0.9040	1.0000	1.0000
TGFB	14/393	3.6%	19/841	2.3%	1.58 (0.80-3.11)	0.1894	0.9468	0.3156
Nrf2	11/393	2.8%	24/841	2.9%	0.98 (0.49-1.98)	1.0000	1.0000	1.0000